

Ahnaf Rafi Sharan

📍 Pabna-6600, Bangladesh ✉ ahnaf@ce.uui.ac.bd ☎ +8801796549094

🌐 [linkedin.com/in/ahnaf-rafi-sharan-06a0391b3](https://www.linkedin.com/in/ahnaf-rafi-sharan-06a0391b3) 📄 scholar.google.com/citations

EDUCATION

Master of Science (M. Sc.) in Civil and Environmental Engineering

Islamic University of Technology (IUT)

CGPA (Most Recent Semester): **3.75/4.00**

11/2025 – Present

Gazipur, Bangladesh

Bachelor of Science (B. Sc.) in Civil and Environmental Engineering

Islamic University of Technology (IUT)

CGPA: **3.91/4.00** (Rank: 3rd out of 96)

06/2021 – 10/2025

Gazipur, Bangladesh

RESEARCH INTERESTS

- Multimodal Traffic Data Analytics
- AI-Based Traffic Flow and Road Safety Analysis
- Digital Twins for Transportation Infrastructure
- Intelligent Transportation Systems (ITS)
- Transportation Policy and Planning
- Highway & Airport Pavement Engineering

RESEARCH EXPERIENCE

Developing Crash Prediction Model for Highways Considering Land Use and Encroachment (Undergraduate Thesis)

Supervisor: Moinul Hossain, Ph.D., Professor, CEE, IUT

- Compiled and processed crash data from the Dhaka–Sylhet National Highway (N2) and performed feature selection for crash prediction modeling.
- Developed and validated a Negative Binomial Regression model, with spatial and statistical visualization of crash-risk patterns.

Impact of unauthorized vehicles on road safety using machine learning in Bangladesh

Supervisor: Dr. Md. Rashedul Haque, Professor, Department of Civil Engineering, Pabna University of Science and Technology (PUST)

- Defined and categorized unauthorized vehicles in the Bangladesh context based on relevant legal and regulatory frameworks.
- Performed data sorting, cleaning, exploratory analysis, machine learning modeling, and visualization to investigate road safety implications.

Developing spatial crash prediction model for highways considering land use & encroachment

Supervisor: Moinul Hossain, Ph.D., Professor, CEE, IUT; Md. Asif Raihan, Ph.D., Professor, Accident Research Institute (ARI), BUET

- Applied QGIS-based geospatial analysis to examine crash interactions between adjacent highway segments and mapped roadside land use and encroachment patterns.
- Contributed to journal manuscript preparation and developed the research poster for presentation at the 105th TRB Annual Meeting (TRB 2026)

Understanding Crash Risk on Highways: The Impact of Land Use, Encroachment and Built Environment

Supervisor: Moinul Hossain, Ph.D., Professor, CEE, IUT; Md. Asif Raihan, Ph.D., Professor, ARI, BUET

- Investigated the interaction between land use, roadside encroachment, and built-environment characteristics and their influence on highway crash risk.
- Contributed to research manuscript development and designed the conference poster for presentation at the 16th EASTS Conference.

Utilization of recycled brick aggregate to ensure circular economy

Supervisor: Dr. Md. Tarek Uddin, PEng, Professor, CEE, IUT

- Developed predictive models for concrete compressive strength using Multivariable Linear Regression (MVLRL) and Multivariable Full Quadratic (MVFQ) regression.
- Supported model development through statistical analysis and evaluation of influential concrete mix parameters.

Prediction of strength properties of soft soil considering simple soil parameters

Supervisor: Mozaher Ul Kabir Mahadi, Ph.D., Research Assistant, University of Kansas

- Compiled, sorted, and analyzed soft-soil data from 250 boreholes across Dhaka City to investigate relationships between basic soil parameters and strength properties.
- Prepared the complete journal manuscript and developed responses to peer-review comments and rebuttal requirements.

JOURNAL PUBLICATIONS

Haque, M. R., Hossain, M. N., **Sharan, A. R.**, Aman, N., Miah, A. S. M., Istiqphara, S., & Ferdous, M. F. (2026). [Impact of unauthorized vehicles on road safety using machine learning in Bangladesh](#). [International Journal of Engineering](#), 27/04/2026, e-ISSN 1735-9244.

Hoque, M. J., Bayezid, M., **Sharan, A. R.**, Kabir, M. U., & Tareque, T. (2023). *Prediction of strength properties of soft soil considering simple soil parameters*. [Open Journal of Civil Engineering](#), 13, 479–496. <https://doi.org/10.4236/ojce.2023.133035>

CONFERENCE PROCEEDINGS

Uddin, N., Tasnim, A., **Rafi, A.**, Raihan, M. A., Hossain, M., Fletcher, J., & Rahman, A. S. (2026). *Developing spatial crash prediction model for highways considering land use & encroachment*. Paper presented at [the 105th TRB Annual Meeting \(TRB 2026\)](#), Washington, D.C., USA, January 11–15, 2026.

Uddin, N., Tasnim, A., **Rafi, A.**, Raihan, M. A., & Hossain, M. (2025). [Understanding Crash Risk on Highways: The Impact of Land Use, Encroachment and Built Environment](#) . Paper presented at the **16th International Conference of Eastern Asia Society for Transportation Studies (EASTS 2025)**, Surakarta, Indonesia.

Mohammed, T. U., Uddin, N., & **Rafi, A.** (2024). [Utilization of recycled brick aggregate to ensure circular economy](#) . In Proceedings of the 1st International Conference on Core Engineering & Technology (IUT-ICCET 2024), Gazipur, Bangladesh. **IUT Journal of Engineering and Technology (JET)**, *Special Issue*, 848–854.

PROFESSIONAL EXPERIENCE

Lecturer, Department of Civil Engineering <i>United International University (UIU)</i> Relevant Courses Taught: GIS and Remote Sensing Lab, Engineering Hydrology, Engineering Geology and Geomorphology	11/2025 – Present Dhaka, Bangladesh
Industrial Training at SMEC (Bangladesh) Ltd.	03/2024 – 03/2024 Dhaka, Bangladesh

ACADEMIC PROJECTS

Final Year Design Project (FYDP) <i>Design and Planning of an Eight-Story Residential Building in Kishoreganj</i> - Led the integrated design covering geotechnical, structural, environmental, and transportation aspects. - Conducted mat and pile foundation analysis using PLAXIS 3D, selecting piles based on settlement and bearing capacity. - Performed ETABS analysis and design of beams, columns, slabs, and stairs with serviceability and reinforcement checks. - Conducted concrete mix design, service-life prediction, water demand analysis, and rainwater harvesting design using E-Tank. - Conducted Traffic Impact Assessment and SUMO simulation to evaluate future traffic conditions. - Prepared environmental assessment, EMP/EMoP, water and waste management plans, and pavement design.	
--	--

RELEVANT COURSEWORK

- Transportation and Traffic Engineering - GIS Application in Civil Engineering - Transportation Projects and Operations - Probability and Statistics - Computer Programming and Applications	- Civil Engineering Data Analysis - Public Transportation System - Transportation Engineering Sessional I & II - Numerical Methods and Computer Programming - Pavement Design and Railway Engineering
---	---

SKILLS

Technical Skills Analysis, Simulation and CAD Software: AutoCAD, SUMO, VISSIM, CUBE, QGIS, ETABS, SAP2000, E-Tank, Zotero Programming Language: Python Document Preparation: Microsoft Office (Word, Excel and PowerPoint)	
Language Proficiency English: Overall Band Score 7.5 (IELTS) Bengali: Native	

CERTIFICATIONS

Certification of Accomplishment for completing Industrial Training at SMEC BD Ltd. Certification of Accomplishment for ‘Data Visualization with Matplotlib’ in Datacamp Certification of Accomplishment for ‘Data Visualization with Seaborn’ in Datacamp	
--	--

EXTRACURRICULAR ACTIVITIES

CENNOVATION 2025 Hosted by Department of Civil and Environmental Engineering, IUT Position: Assistant General Secretary	IUT, Gazipur
ACI Students’ Chapters Meet 2025 Hosted by ACI IUT Student Chapter Position: Design Team Lead	IUT, Gazipur
ACI IUT Student Chapter Position: Head of Creative Design	IUT, Gazipur
IUT ITE Student Chapter Position: Technical Executive	IUT, Gazipur

COMMUNITY INVOLVEMENT

Winter Clothes Donation, IUT SIKS <i>Participated in donating winter clothing to underprivileged communities in Gazipur</i>	2023 – 2024 Gazipur, Dhaka
Blood Donor, Badhan Jahangirnagar University Zone <i>Active blood donor and volunteer in awareness campaigns</i>	2021 – Present Savar, Dhaka